Blackmon Lab
Evolutionary Biology

Does Geographic Environment Influence Chromosome Number in Orthoptera?

Bevin Haynes, Dr. Health Blackmon
Department of Biology

TEXAS A&M UNIVERSITY Biology

Background

The karyotype, the number and structure of an organism's chromosomes, is one of the most fundamental features of a genome. Despite over a century of cytogenetic study, the forces that drive chromosome number evolution across the tree of life remain poorly understood (Blackmon & Demuth, 2015; Sylvester et al., 2020). Orthoptera (grasshoppers, katydids, and crickets) is one of the most cytogenetically diverse insect orders, with haploid numbers ranging from n = 4 to n = 28 (Husemann et al., 2022). The most common haploid number in Acrididae is n = 12, driven largely by a conserved X0 sex determination system (Hewitt, 1979). However, South American Melanoplinae — particularly Dichroplus and related genera — show striking karyotypic diversification driven by Robertsonian fusions and neo-sex chromosome systems (Colombo et al., 2005; Castillo et al., 2017).

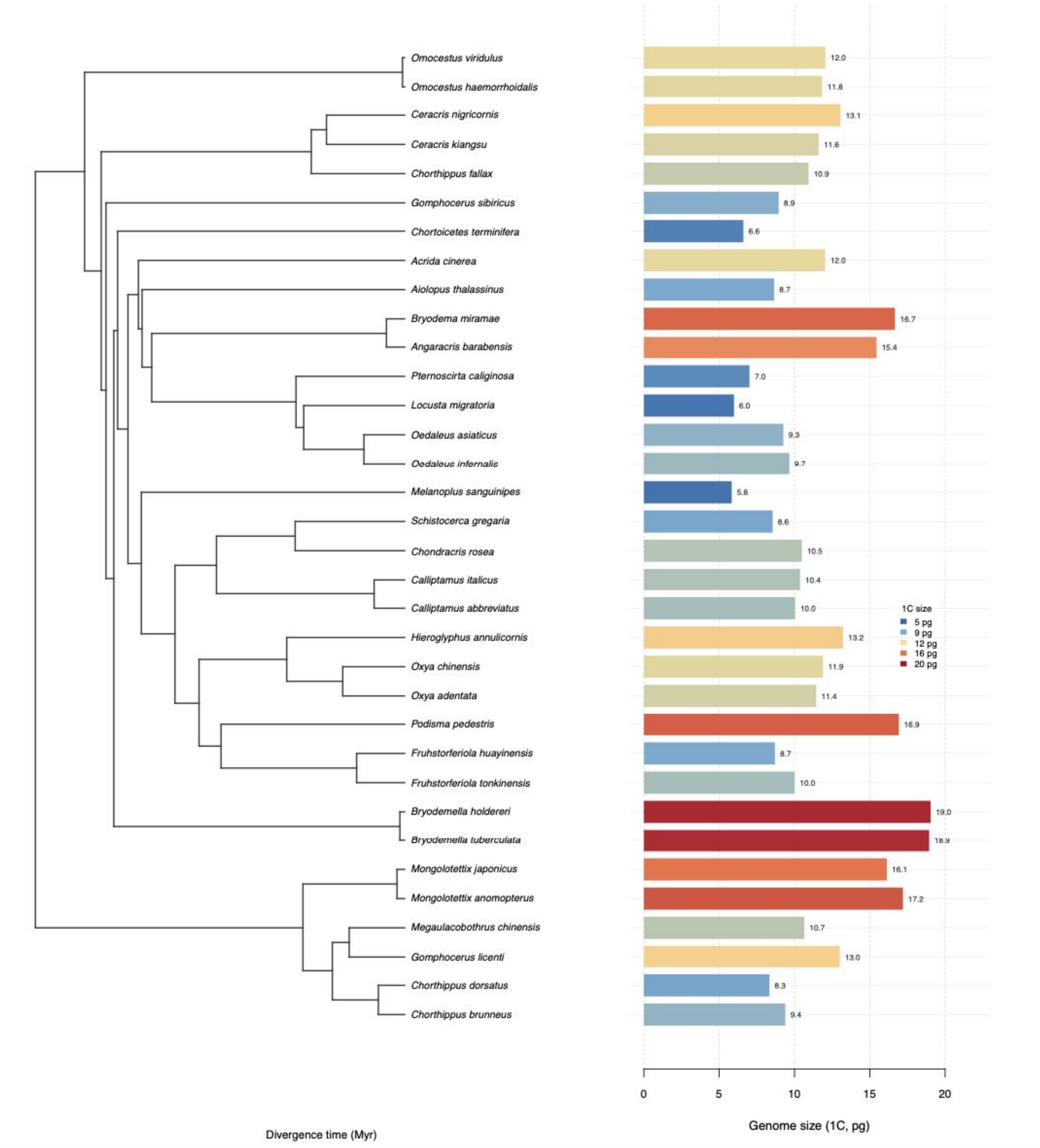
Study Design & AI

Claude assisted in refining the research question from a broad interest in chromosome evolution into a specific, testable hypothesis suitable for a PGLS framework. R scripts for querying GBIF occurrence data, classifying hemisphere from median latitude, retrieving and pruning the Open Tree of Life phylogeny, and running the PGLS model were developed collaboratively with Claude, with iterative debugging of package conflicts and tree formatting issues. R scripts for querying GBIF occurrence data, classifying hemisphere from median latitude, retrieving and pruning the Open Tree of Life phylogeny, and running the PGLS model were developed collaboratively with Claude, with iterative debugging of package conflicts and tree formatting issues. Claude identified that the appropriate method for this discrete-continuous question was PGLS using gls() with corBrownian(), consistent with the Blackmon Lab PCM guide.

Model Fitting

- Haploid Data: Species-level haploid chromosome numbers compiled from the CUREs Karyotype Database (Blackmon Lab, Texas A&M), covering 317 Orthoptera species across multiple families.
- Habitat Classification: Occurrence records (latitude/longitude) retrieved for each species from the Global Biodiversity Information Facility (GBIF) via the rgbif R package. Median latitude was used to classify species as Northern Hemisphere (≥ 0°, coded 1) or Southern Hemisphere (< 0°, coded 0).
- Phylogeny: A species-level synthetic tree was retrieved from the Open Tree of Life (OTL) API via the rotl R package. Branch lengths were assigned using Grafen's method and the tree was made ultrametric with chronos().
- Statistical model: Phylogenetic Generalized Least Squares (PGLS) was fitted using gls() from the nlme package with a Brownian motion correlation structure (corBrownian), following Blackmon Lab methods.

Results



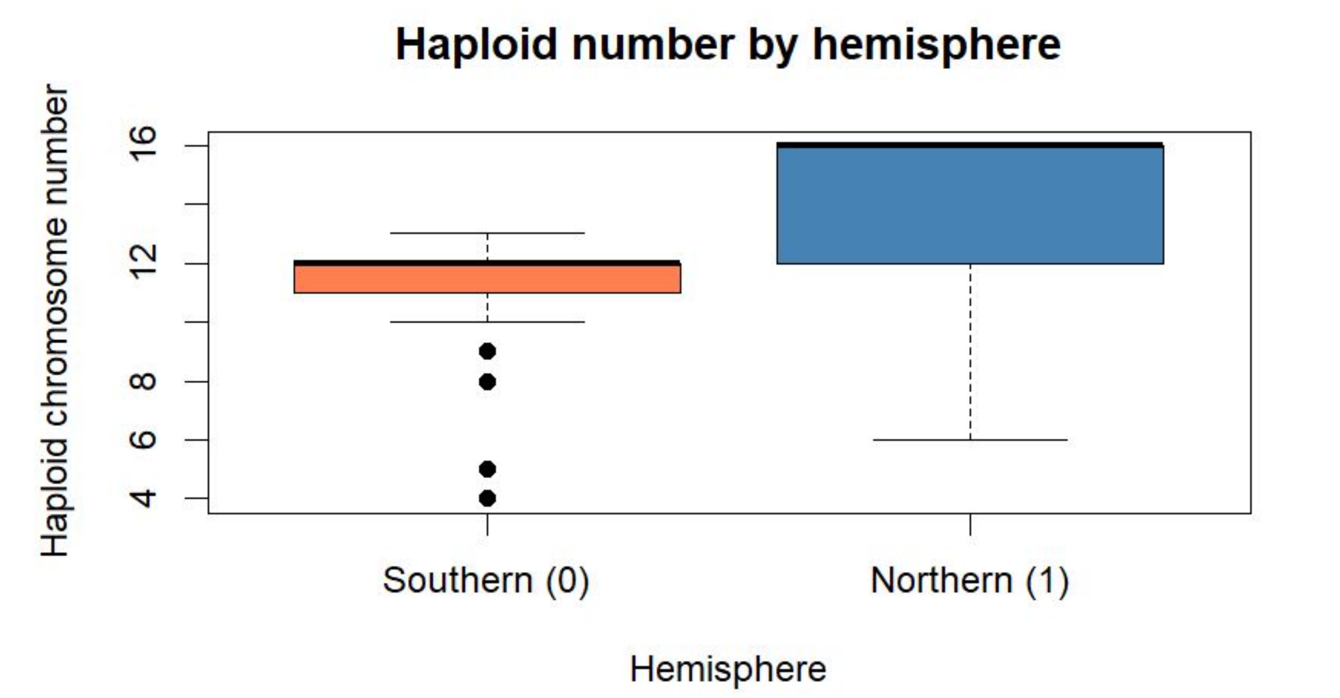
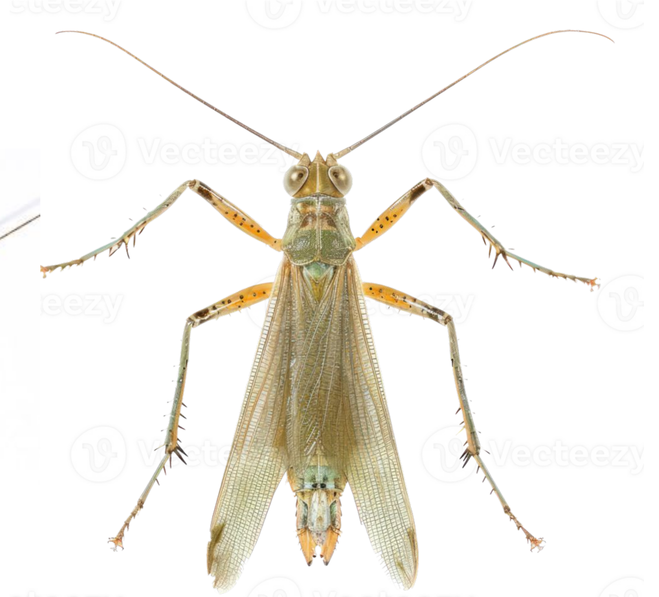


Fig. 1. Haploid chromosome number by hemisphere. Median haploid number is similar between groups (Southern: 11.65 ± 7.29 SE; Northern: 11.71).



Discussion

The PGLS model reveals that northern vs. southern hemisphere classification explains essentially none of the variation in haploid chromosome number (F = 0.024, p = 0.877). The estimated difference between groups is negligible (+0.054 chromosomes), far smaller than the residual standard error of 7.29, which itself reflects the enormous within-group variation in Orthoptera karyotypes. This result is consistent with broader findings in Polyneoptera, where lineage-specific processes — rather than environmental gradients — appear to dominate chromosome number evolution (Sylvester et al., 2020). In Orthoptera specifically, rates of chromosome fusion are among the lowest estimated across Polyneoptera, while polyploidy rates are comparatively elevated. These patterns suggest that internal genomic processes, rather than external environmental pressures, shape karyotype diversity.

The dataset is dominated by South American Melanoplinae (particularly Dichroplus and related genera), a group notable for extreme karyotypic diversification driven by Robertsonian fusions and neo-sex chromosome evolution (Colombo et al., 2005; Castillo et al., 2017). This within-hemisphere variation likely overwhelms any broad-scale hemisphere signal.

<u>Limitations</u>: The dataset is heavily imbalanced (132 southern vs. 14 northern species), which reduces power. Hemisphere is also a coarse proxy for environment. Future work should use finer-grained climate variables (temperature seasonality, precipitation) and a more balanced hemispheric sample.

Conclusions & Acknowledgements

Geographic environment, as operationalized by hemisphere, does not explain variation in haploid chromosome number across Orthoptera. The null hypothesis of no effect cannot be rejected (p = 0.877). Chromosome number evolution in this group appears to be driven primarily by lineage-specific genomic processes such as Robertsonian fusions and neo-sex chromosome formation, consistent with the broader comparative literature on insect karyotype evolution. Finer-grained environmental variables and a more taxonomically balanced dataset will be necessary to fully evaluate the role of ecology in shaping Orthoptera genome structure. Special thanks to Dr. Heath Blackmon (Texas A&M University) for guidance on phylogenetic comparative methods and access to the CUREs Karyotype Database. This project used the Blackmon Lab PCM guide (coleoguy.github.io/phylo-methods) as a methodological reference. Occurrence data were provided by the Global Biodiversity Information Facility (GBIF). The phylogenetic framework was made possible by the Open Tree of Life project.